
CLINTON ENWEREM

Academic CV

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RESEARCH INTERESTS

Dexterous Robotic Grasping & Manipulation, Risk-Sensitive Planning & Control, Safety-Critical Control, Robot Learning, Multi-Modal Uncertainty Estimation.

EDUCATION

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| 2021–Fall 2026 | Ph.D., Electrical & Computer Engineering <i>University of Maryland, College Park</i> MD, USA
Advisors: Professor John S. Baras and Professor Calin Belta . Admitted to Candidacy: June 2025.
<i>Relevant Coursework:</i> Decision-Making for Robotics, Network Control Systems, Decision Making Under Uncertainty, Random Processes, Advanced Digital Signal Processing, Nonlinear Control Systems, Optimal Control, System Theory, Convex Optimization, Formal Methods for Control & Dynamical Systems, Multimodal Foundational Models. |
| 2014–2018 | B.Eng., Electrical Engineering <i>University of Nigeria, Nsukka</i> Enugu, Nigeria
<i>Graduated with Highest Honors. Emphasis:</i> Control Theory. <i>Completed coursework in engineering numerical analysis, control systems, electrical machines and drives, computer architecture, electronics, and communication systems.</i> |

PUBLICATIONS

Preprints / Articles In Review

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| 2026 | [P3] Clinton Enwerem , John S. Baras, and Calin Belta. FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis, July 2026. arXiv Link . |
| 2026 | [P2] Clinton Enwerem , John S. Baras, and Calin Belta. EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows, June 2026. arXiv Link . |
| 2026 | [P1] Clinton Enwerem , John S. Baras, and Calin Belta. Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality. Manuscript, 2026. Project Page . |

Refereed Conference Publications

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| 2026 | [C6] Clinton Enwerem , John S. Baras, and Calin Belta. Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty, December 2026. <i>To appear in the Proceedings of the 65th IEEE Conference on Decision and Control (CDC)</i> . arXiv Link . |
| 2026 | [C5] Clinton Enwerem , Shreya Kalyanaraman, John S. Baras, and Calin Belta. Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty, October 2026. <i>To appear in the Proceedings of the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> . arXiv Link . |
| 2025 | [C4] Clinton Enwerem , Aniruddh G. Puranic, John S. Baras, and Calin Belta. Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression. <i>2025 IEEE 64th Conference on Decision and Control (CDC)</i> , pages 4890–4895, December 2025. arXiv Link . |
| 2024 | [C3] Clinton Enwerem , Erfan Noorani, John S. Baras, and Brian M. Sadler. Robust Stochastic Shortest-Path Planning via Risk-Sensitive Incremental Sampling. <i>2024 IEEE 63rd Conference on Decision and Control (CDC)</i> , pages 1087–1094, December 2024. arXiv Link . |
| 2024 | [C2] Clinton Enwerem and John S. Baras. Safe Collective Control under Noisy Inputs and Competing Constraints via Non-Smooth Barrier Functions. <i>2024 European Control Conference (ECC)</i> , pages 3762–3768, June 2024. |
| 2023 | [C1] Clinton Enwerem and John S. Baras. Consensus-Based Leader-Follower Formation Tracking for Control-Affine Nonlinear Multiagent Systems. <i>9th International Conference on Control, Decision and Information Technologies (CoDIT)</i> , July 2023. |

Refereed Journal Articles

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| 2024 | [J1] Clinton Enwerem and John S. Baras. Formation Tracking for a Class of Uncertain Multiagent Systems: A Distributed Kalman Filtering Approach. <i>IEEE Control Systems Letters</i> , volume 8, 2024. |
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HONORS & AWARDS

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| 2024 | IEEE CSS Student Travel and Workshop Support: Conference travel award to attend CDC’24. |
| 2022 | ‘22–‘23 Microsoft Diversity in Robotics & Autonomy PhD Fellowship, MRC & Microsoft. |

2022	ROSCon Diversity Scholarship: Travel grant to attend ROSCon 2022 in Kyoto, Japan.
2021	Finalist, Engineers' League, Pan-African Robotics Competition, Rwanda.
2021	CIT Dean's Fellowship, Carnegie Mellon University, Africa Campus, Kigali, Rwanda.
2021	Dean's Fellowship, University of Maryland, College Park, MD, United States.
2020	Scholar, Stanford Exposure to Research & Graduate Education, Stanford University, CA, USA.
2020	EducationUSA Opportunity Funds Program Scholarship, U.S. Consulate General, Lagos, Nigeria.
2020	Sole Recipient (Nationwide), Door Foundation Leadlight Scholarship, RAIN.
2016–2018	Agbami Science & Technology Scholarship, Chevron: Merit-based undergraduate scholarship.
2015–2018	MTN Foundation Scholarship: Nationwide merit-based undergraduate scholarship.

RESEARCH EXPERIENCE

8/2021–Present Research Topics Duties	Graduate Research Assistant <i>Institute for Systems Research, University of Maryland</i> <i>Learning-Enabled Methods for Robust Dexterous Robotic Grasping & Manipulation</i> - Design learning-enabled grasping and manipulation algorithms for robotic arms with anthropomorphic end-effectors, spanning perception, belief estimation, planning, and control. - Derive risk-sensitive grasp quality metrics with probabilistic force-closure certificates, and build the certified execution controllers that hold those margins through contact. - Develop and maintain performant and research-grade software in Python and C++ for grasp synthesis, stochastic motion planning, and closed-loop manipulation control, deployed on fixed-base arms and on a 21-DOF fixed-pelvis humanoid whole-body chain in simulation. - Benchmark algorithms over large object sets in MuJoCo and Drake, and validate learned policies in OpenAI Gym and Safety Gymnasium across contact-rich manipulation tasks. - Disseminate findings through peer-reviewed conference and journal papers, with work published at CoDIT, ECC, CDC, and IEEE L-CSS, accepted at IROS 2026 and CDC 2026, and posted as arXiv preprints.	College Park, MD
9/2018–3/2021 Research Topics Accomplishments	Research Assistant <i>Electrical Engineering Department, University of Nigeria</i> <i>Robust Control, Observer-Based Compensator Design, Feedback Linearization</i> - Designed and implemented robust nonlinear control, active disturbance rejection, feedback linearization, and observer-based compensation for hardware motor control, released as open-source software. - Co-authored the resulting peer-reviewed journal paper.	Enugu, Nigeria
8/2017–10/2017 Research Topics Accomplishments	Undergraduate Research Assistant <i>EE Department, University of Nigeria</i> <i>Feedback Control, Time-Delayed Systems, System Identification</i> Designed feedback controllers for first-order-plus-dead-time processes with dead-time compensation and closed-loop system identification; implementation and write-up available online.	Enugu, Nigeria

PROFESSIONAL EXPERIENCE

Jun.–Aug. 2023 Supervisor Accomplishments	Summer Research Assistant <i>Institute for Systems Research, University of Maryland</i> <i>Professor John S. Baras</i> - Formulated a multi-agent safety-critical control problem as a chance-constrained mathematical program. - Proposed a solution based on Boolean-composed control barrier certificates. - Wrote software to validate the approach and wrote up the results as a research paper.	College Park, MD
Jun.–Aug. 2022 Supervisor Accomplishments	Research Intern <i>MATRIX Lab, USM at Southern Maryland</i> <i>Dr. Danilo Romero</i> - Ran system identification experiments to validate a quadrotor model. - Developed a Lagrangian-based optimal swarm control algorithm coordinating 10 Crazyflie quadrotors for formation tracking under localization uncertainty, and wrote the ROS-compliant Python implementation.	California, MD
3/2020–2/2021 Accomplishments	Robotics Engineer <i>Kognitive Robotics</i> <i>Kognitive Robotics is a robotics startup developing turnkey robotics education platforms.</i> - Brought up a mobile robot end to end, designing its power and sensor-actuator interface electronics (Altium schematic capture and PCB layout) and integrated the sensing and actuation stack. - Developed the robot's ROS-based software stack and validated hardware-software integration on the physical platform.	Lagos, Nigeria

1/2020–2/2021
Accomplishments

Robotics Trainee | *Robotics & Artificial Intelligence Nigeria (RAIN)*

Ibadan, Nigeria

- Contributed to robotics and IoT projects across the hardware and software stack, spanning computer-aided design, rapid prototyping, sensor fusion, control firmware, and product testing.
- Built ROS-based visual-SLAM, state-estimation, and flight-control software for a low-cost UAV on embedded hardware.

TEACHING EXPERIENCE

9/2025–12/2025
Course
Accomplishments

Graduate Teaching Assistant | *ECE Department, University of Maryland*

College Park, MD

ENEE467: Robotics Project Laboratory (with Prof. Calin Belta)

- Mentored 50+ undergraduate students in kinematics, 3D perception, motion planning, and ROS 2 control.
- Built reproducible Docker-based labs and testing pipelines.
- Led weekly lab sessions and provided technical feedback on documentation and debugging.

2/2025–5/2025
Course
Accomplishments

Graduate Teaching Fellow | *ECE Department, University of Maryland*

College Park, MD

ENEE661: Nonlinear Control Systems (with Prof. John S. Baras)

- Led problem-solving sessions on nonlinear analysis and control, covering phase-plane and bifurcation analysis, and stability concepts including Lyapunov, input-to-state, L_2 , and L_∞ stability.

TECHNICAL SKILLS

Robotics: ROS/ROS2, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, RViz, MoveIt2; **Control & Safe Learning:** Safe & Constrained Reinforcement Learning, Control Barrier Functions, Safety Filters, Constrained MDPs, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Lyapunov Stability, Sim-to-Real Transfer; **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash, $\text{\LaTeX}$; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi, Mosek, Pyomo; **RL Sandboxes:** OpenAI Gym, Safety Gymnasium, PyBullet; **Dev Tools:** git, GitHub, GitLab, Docker; **Robots & EoATs:** Fairino FR3, xArm7, UR3e/UR5, Unitree G1 (simulation), LEAP Hand v1, RealHand L6.

PROFESSIONAL TRAINING & DEVELOPMENT

1/2020–2/2021

Certificate in Robot Development & Automation | *RAIN*

Completed graduate-level coursework and projects in robotics, control theory, ML, and IoT.

S'22/'23/'25/F'25

Open Courseware: Bayesian Statistics (*UCSD, Coursera*), Autonomous Navigation for Flying Robots (*TUM, edX*), Principles of Robot Autonomy I & II (*Stanford*), Planning & Learning in Robotics (*UCSD*), Robot Learning (*USC*).

PROFESSIONAL SERVICE, OUTREACH, & MENTORING

2023–2026

Peer Reviewer: Heliyon, MED'23, ECC'24, ACC'25, L4DC'25, CDC'25, CDC'26, Elsevier (Robotics & Autonomous Systems).

2021–Present

Member, Black in Robotics (BiR): the first organization of its kind, advancing Black representation across academic and industry robotics.

2021–2022

Mentor, EducationUSA & iScholar Initiative: Guided STEM graduates from underrepresented backgrounds to secure fully-funded graduate admissions at Stanford, Purdue, and Penn State.